

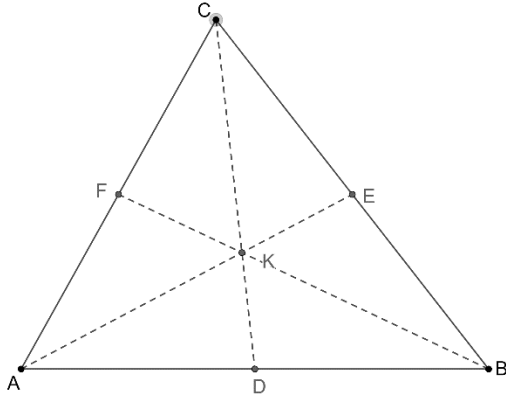
Geometry
Unit 11
Lesson 7 Practice

Name **Answer Key**
Date _____
Period _____

$\triangle ABC$ is shown. Complete the steps to for finding the centroid of the circle. Then, label the centroid Point G .

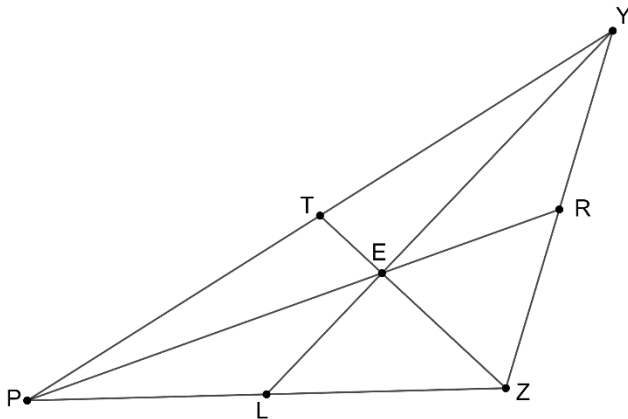
Steps	Construction
1. Construct the perpendicular bisector of $\overline{AB}$ to find its midpoint, R .	5.
Draw the median, R , to the opposite vertex, $\angle ACB$.	
2. Construct the perpendicular bisector of $\overline{BC}$ to find its midpoint, T .	
Draw the median, T , to the opposite vertex, $\angle BAC$.	
3. Construct the perpendicular bisector of $\overline{AC}$ to find its midpoint, S .	
4. Draw the median from Point S to the opposite vertex, $\angle ABC$.	
Draw Point G , the point of concurrency, where the three medians intersect.	

Triangle ABC is shown, where point K is the centroid of the triangle. $FB = 7.5$, $KC = 4.5$, and $KA = 4.8$. Use this information to answer questions 6 through 8.



6. Find the length of $\overline{CD}$. $\frac{2}{3}CD = 4.5$
 $CD = 6.75$
7. Find the length of $\overline{KE}$. $2KE = 4.8$
 $KE = 2.4$
8. Find the length of $\overline{BK}$. $\frac{2}{3}(7.5) = BK = 5$

Triangle YPZ is shown, where point E is the centroid of the triangle. $PE = 3x + 8$, and $RE = 2x - 4$. Use this information to answer questions 9 and 10.



9. Find the value of x . $2(3x - 4) = 3x + 8$; $4x - 8 = 3x + 8$; $x = 16$
10. Find the length of $\overline{PR}$. $(3(16) + 8) + (2(16) - 4) = 84 = PR$